

Deliverable 10 – Business Insights Report

Healthcare Patient Experience Analytics using Power BI

Introduction

The Healthcare Patient Experience Analytics dashboard was developed to evaluate patient satisfaction across national, regional, and state levels using HCAHPS survey data. The dashboard enables healthcare leaders to identify performance trends, benchmark healthcare quality, and support evidence-based decision-making.

The following insights were derived from the interactive Power BI dashboards developed during this project.

Business Insights

No.	Observation	Business Impact	Recommendation
1	The overall national patient satisfaction score is 71.41% .	Patient satisfaction is generally positive but still leaves room for improvement.	Focus on improving low-performing quality measures to increase the overall satisfaction score.
2	The overall survey response rate is 26.61% , indicating relatively low patient participation.	Low participation may reduce the reliability of survey-based decisions.	Encourage more patients to complete surveys through awareness campaigns and digital feedback systems.
3	A total of 51 states and jurisdictions participated in the survey.	The dataset provides broad geographical coverage for performance analysis.	Continue maintaining nationwide participation for consistent benchmarking.
4	The survey evaluates 10 healthcare quality measures .	Multiple quality dimensions provide a comprehensive evaluation of patient experience.	Continue monitoring all quality measures to ensure balanced healthcare improvement.
5	Discharge Information achieved the highest	Hospitals perform exceptionally well in	Adopt discharge communication practices as

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6	patient satisfaction score (86.67%). Care Transition recorded the lowest satisfaction score (52.56%).	providing discharge instructions. Weak care transition processes may negatively affect patient recovery and continuity of care.	best practices across other healthcare services. Improve discharge planning, follow-up communication, and coordination between healthcare providers.
7	National patient satisfaction steadily improved from 2015 to 2021 before declining in 2022 and 2023 .	Recent declines indicate emerging challenges affecting patient experience.	Investigate factors contributing to the decline and implement corrective quality improvement initiatives.
8	The West North Central region consistently demonstrates the highest regional satisfaction scores.	The region represents a benchmark for healthcare quality.	Study successful practices implemented within this region and replicate them elsewhere.
9	The Mid-Atlantic region consistently reports the lowest patient satisfaction.	Regional disparities indicate unequal healthcare service quality.	Prioritize targeted quality improvement programs within the Mid-Atlantic region.
10	States such as South Dakota, Nebraska, and Louisiana appear among the highest-performing states.	These states demonstrate strong healthcare service quality and patient satisfaction.	Identify operational best practices and share them with lower-performing states.
11	The District of Columbia records the lowest overall satisfaction score.	Persistent low performance indicates significant opportunities for improvement.	Conduct detailed root cause analysis and implement targeted quality improvement initiatives.
12	Communication with Doctors (81.33%) and Communication with Nurses (80.11%) receive consistently high patient satisfaction.	Strong communication contributes positively to patient experience.	Continue staff communication training and maintain current service standards.
13	Hospital Cleanliness achieved a satisfaction score above 74% .	Cleanliness positively influences patient confidence and hospital reputation.	Continue infection control and housekeeping quality programs.
14	Responsiveness of Hospital Staff scores below several communication-	Delayed response to patient requests may reduce overall satisfaction.	Improve staffing efficiency and reduce patient response times.

No.	Observation	Business Impact	Recommendation
	related measures.		
15	Communication about Medicines remains comparatively lower than doctor and nurse communication.	Patients may not fully understand prescribed medications and treatment plans.	Strengthen medication counseling before patient discharge.
16	Willingness to Recommend the Hospital closely follows the overall patient satisfaction trend.	Better patient experience directly improves hospital reputation.	Continue focusing on patient-centered healthcare services.
17	National vs State Benchmark analysis shows a variance of - 0.08 , indicating state performance is slightly below the national average.	Small performance gaps still affect overall healthcare quality.	Focus improvement efforts on states performing below national benchmarks.
18	Several healthcare measures outperform the national average, while Quietness of Hospital Environment and Overall Hospital Rating remain below benchmark.	Performance variation exists across different quality measures.	Prioritize investment in hospital environment improvements.
19	Response rate trends gradually decline over the reporting periods.	Lower participation reduces the representativeness of patient feedback.	Introduce digital survey collection methods and patient engagement initiatives.
20	Year-over-Year Satisfaction Change is -1.30 , indicating an overall decline in recent reporting periods.	Declining satisfaction may affect healthcare quality ratings.	Monitor satisfaction trends regularly and intervene before performance deteriorates further.
21	Regional performance varies significantly across the United States.	Regional differences indicate inconsistent healthcare delivery standards.	Establish standardized quality improvement programs across all regions.
22	Interactive dashboards enable executives to compare states, regions, measures, and	Faster access to healthcare insights supports better strategic decisions.	Continue using Business Intelligence dashboards for executive performance monitoring.

No.	Observation	Business Impact	Recommendation
	reporting periods in real time.		
23	KPI-driven reporting simplifies performance evaluation across multiple healthcare dimensions.	Decision-makers can quickly identify strengths and weaknesses.	Expand KPI monitoring to additional healthcare quality indicators in future reporting.
24	Benchmark analysis clearly highlights performance gaps between healthcare measures.	Gap analysis helps prioritize quality improvement initiatives.	Focus resources on measures with the largest negative variance.
25	The Power BI solution provides a centralized platform for monitoring patient experience nationwide.	Interactive reporting improves transparency, collaboration, and evidence-based decision-making.	Integrate additional healthcare datasets to enhance future analytical capabilities.

Conclusion

The analysis reveals that healthcare performance across the United States is generally positive, with an overall patient satisfaction score above 70%. However, noticeable differences exist between regions, states, and healthcare quality measures. While **Discharge Information** consistently demonstrates excellent performance, **Care Transition** remains the weakest-performing measure and requires immediate attention.

The dashboard also indicates declining patient satisfaction and response rates in recent years, emphasizing the need for continuous monitoring and targeted quality improvement initiatives. By leveraging interactive Power BI dashboards and KPI-driven reporting, healthcare stakeholders can make informed decisions, prioritize improvement efforts, and enhance the overall patient experience.

